

ViolationLens AI

Automated Photo Identification and Classification for Traffic Violations Using Computer Vision

From raw traffic photo to review-ready violation evidence in seconds.

Submission Type: Concept Note / Prototype Proposal / Solution Framework

Theme: Automated Photo Identification and Classification for Traffic Violations Using Computer Vision

Positioning: ViolationLens AI is an evidence-ready traffic violation intelligence platform that automatically analyzes traffic images, enhances image quality, detects vehicles and road users, classifies violations, extracts license plate details, generates annotated evidence, stores metadata, and supports human review and analytics for scalable enforcement.

Prototype Capability	Implementation Status
Image upload and preprocessing	Implemented in backend pipeline using OpenCV quality scoring, enhancement and storage.
Vehicle and road-user detection	YOLOv8-based detection with custom model plug-in readiness.
Violation classification	Rule-engine based MVP for helmet, triple riding, red-light, stop-line, illegal parking, wrong-side suspected and seatbelt model-ready cases.
License plate OCR	OCR fallback with confidence and review warnings.
Evidence generation	Annotated image output, metadata JSON, timestamp and evidence hash.
Human review and analytics	Review queue, approval workflow, summary KPIs, trends and evaluation endpoint.

1. Problem Statement

Traffic surveillance systems generate a large volume of photographic evidence every day. Manual inspection of these images is slow, labour-intensive and inconsistent, especially when images are affected by low light, rain, shadows, motion blur, occlusions or crowded traffic. Enforcement teams need a scalable system that can automatically identify and classify violations while preserving a human review layer for final decisions.

2. Proposed Solution

ViolationLens AI is a computer vision evidence-generation platform. It accepts a traffic image, captures scene context such as signal status, road direction, weather, camera angle and stop-line position, preprocesses the image, detects vehicles and road users, runs OCR on plate regions, applies configurable violation rules, generates annotated evidence and stores the complete record for review and analytics.

3. Innovation and Differentiation

- Evidence-first workflow: the output is not just a prediction but a review-ready evidence record with annotated images, metadata and hash.
- Scene-aware violation logic: still images are combined with user-configured signal status, stop-line position, road direction, weather and camera angle.
- Responsible AI: all AI outputs are routed to human review before enforcement action.
- Safe fallback design: missing custom model files, low-quality images and OCR failures do not crash the system; they create clear warnings.
- Scalable architecture: frontend pages are backed by modular FastAPI endpoints, PostgreSQL records and pluggable YOLO model weights.

4. System Architecture

The architecture follows a modular flow from image upload to evidence storage and analytics. The same flow can later be scaled using GPU inference, queue-based processing, S3 evidence storage and camera-wise calibration profiles.



5. AI Pipeline

Image Preprocessing

Brightness score, blur score, CLAHE enhancement for low light, denoising, sharpening and image quality labelling.

Vehicle and Road User Detection

YOLOv8 detects common traffic objects such as car, motorcycle, bus, truck, bicycle and person. Custom .pt files can be plugged in for helmet, seatbelt, number plate or illegal-parking-specific classes.

License Plate OCR

Vehicle or plate regions are cropped and OCR is applied. The result includes plate text, confidence and status. Failed OCR does not block evidence generation.

Violation Rule Engine

Rules combine detections with scene context and geometry: stop-line crossing, red signal, person-motorcycle overlap and no-parking polygon checks.

Evidence Generation

The system produces annotated images with bounding boxes, labels, confidence values, violation reasons, timestamp and evidence hash.

6. Violation Detection Logic

Violation	Prototype Logic	Review Strategy
Helmet non-compliance	Motorcycle/rider detected and helmet absent or no-helmet model fires.	Manual review if confidence is low or helmet is occluded.
Seatbelt non-compliance	Car/driver region custom model-ready; low visibility routed to review.	Review required until trained model confidence is validated.
Triple riding	Three or more persons overlap with one motorcycle region.	Officer confirms rider grouping.
Wrong-side driving	Still image is marked suspected unless movement/tracking exists.	Responsible review-only classification for still images.
Stop-line violation	Vehicle front point crosses configured stop_line_y.	Review when stop line is unclear.
Red-light violation	Signal status is RED and vehicle crosses stop line.	Requires signal context and officer confirmation.
Illegal parking	Vehicle bottom-center point inside no-parking polygon.	Officer confirms zone and vehicle position.

7. Analytics and Reporting

The dashboard and analytics pages are driven from PostgreSQL evidence records. Operational metrics can be calculated accurately even in the prototype stage: total images processed, pending review count, approved and rejected cases, average confidence, OCR success rate, evidence generated count, low-light cases, violation count by type, location-wise counts and average processing time.

Suggested dashboard KPIs:

- Total violations processed
- Pending review queue
- Approved and rejected cases
- Average confidence
- OCR success rate
- Average processing time
- AI reliability score = average confidence × approval rate
- High-risk locations by violation count

8. Performance Evaluation

For the prototype, operational performance is measured from actual records: image processing time, evidence generation success rate, OCR success rate and average confidence. Final ML metrics such as accuracy, precision, recall, F1-score and mAP require a labelled ground-truth test set. The downloaded YOLO-format datasets with train/test image and label folders can be used for model training and proper evaluation after submission.

9. Dataset and Training Plan

The current submission can show dataset readiness by including screenshots or references to downloaded helmet, rider, number plate and related traffic datasets. The backend is custom-model ready: trained weights such as helmet_best.pt, seatbelt_best.pt and number_plate_best.pt can be placed under backend/models and enabled without changing frontend routes or API contracts.

Immediate prototype use:

- Use selected dataset images as demo_samples and upload them through the AI Analysis Studio.
- Do not train during the final deadline window; training can be highlighted as the next implementation milestone.
- Use pretrained YOLOv8 plus rule engine plus OCR fallback to demonstrate the end-to-end evidence workflow.

10. Edge-case Handling

Edge Case	System Handling
Low light	CLAHE enhancement and low-light warning.
Motion blur	Blur score, sharpening and manual review when

	poor.
Rain/noise	Denoising and confidence warning.
Plate unreadable	OCR failure stored while evidence record is still created.
Missing custom model	Warning returned; pipeline does not crash.
Unknown signal or road direction	Related violation marked suspected or skipped for safe review.
No stop line configured	Stop-line and red-light rules are skipped safely.

11. Deployment and Submission Plan

- Deploy frontend prototype on Vercel.
- Upload backend and frontend code to GitHub.
- Provide concept note, PPT deck and system architecture diagram.
- Provide short demo recording showing upload, analysis, evidence output, records, review and analytics.
- Provide dataset proof through folder screenshots or Kaggle source references.

12. Expected Outcome

ViolationLens AI demonstrates a scalable AI-based traffic image analysis system capable of identifying, classifying and documenting traffic violations from photographic evidence. The prototype reduces manual inspection effort, improves evidence traceability and provides a practical path toward production-grade enforcement analytics.